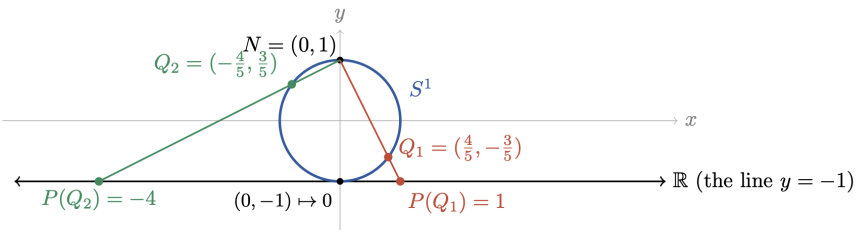
Topology
Home Work 5

Student: 26-712-224

October 15, 2026

Exercise 1. Let $\mathbb{S}^1$ denote the unit circle (equipped with the topology induced from $\mathbb{R}^2$), let $\mathbb{R}_\infty = \mathbb{R} \cup \{\infty\}$ and let $P : \mathbb{S}^1 \to \mathbb{R}_\infty$ be the map defined by

$$P(x,y) = \begin{cases} \dfrac{2x}{1-y}, & \text{if } (x,y) \neq (0,1), \\ \infty, & \text{if } (x,y) = (0,1). \end{cases}$$



This map is known as stereographic projection. Let $\mathcal{U}_{\mathbb{R}_\infty}$ denote the quotient topology on $\mathbb{R}_\infty$ induced by P.

(i) Prove that for all $a < b \in \mathbb{R}$, $(a,b) \in \mathcal{U}_{\mathbb{R}_\infty}$.

(ii) Let $\mathcal{B}_\infty = \{(-\infty, a) \cup (b, +\infty) \cup \{\infty\} : a < b \in \mathbb{R}\}$, and prove that $\mathcal{B}_\infty$ forms a base of open neighborhoods of the point ∞.

(iii) Prove that $\{(a,b) : a, b \in \mathbb{R}\} \cup \mathcal{B}_\infty$ form a basis for $\mathcal{U}_{\mathbb{R}_\infty}$. (You can use the face that arcs on $\mathbb{S}^1$ form a base for its topology.)

Exercise 2. The goal of this exercise is to prove that the Hausdorff property is not preserved by quotients. Consider two lines on the plane $L_0 = \{(x, 0); x \in \mathbb{R}\}$, $L_1 = \{(x, 1); x \in \mathbb{R}\}$, and equip $L = L_0 \cup L_1$ with the topology induced by $\mathbb{R}^2$. Now consider the equivalence relation on L defined as

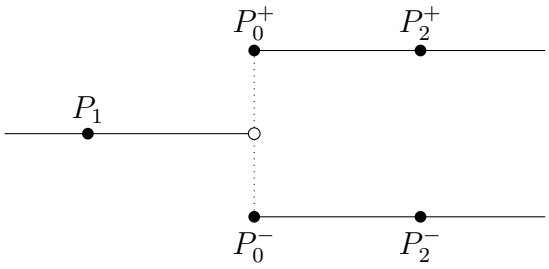
$$(x, 0) \sim (y, 1) \quad \text{iff } x = y < 0$$

and define the space $X = L/\sim$ (with quotient topology), along with the projection $p : L \to X$.

i) Show that L is Hausdorff.

ii) Let $x_1 < 0 < x_2$ and define $P_1 = p(x_1, 0) = p(x_1, 1)$, $P_2^+ = p(x_2, 1)$, $P_2^- = p(x_2, 0)$, $P_0^+ = p(0, 1)$, $P_0^- = p(0, 0)$. Describe the neighbourhoods in X of P_1, P_0^+, P_0^-, P_2^+ and P_2^-.

iii) Prove that X is <u>not</u> Hausdorff.



Exercise 3. The goal of this exercise is to show that two definitions of the torus are topologically equivalent, namely that

$$\mathbb{R}^2/\mathbb{Z}^2 \cong \mathbb{S}^1 \times \mathbb{S}^1.$$

Consider $\mathbb{R}^2/\mathbb{Z}^2$ with the quotient topology arising from the natural group action of $\mathbb{Z}^2$ on $\mathbb{R}^2$: for $(n,m) \in \mathbb{Z}^2$, $(x,y) \in \mathbb{R}^2$, we define $(n,m)\cdot(x,y) = (n+x, m+y)$. Let $p : \mathbb{R}^2 \to \mathbb{R}^2/\mathbb{Z}^2$ be the quotient map. The topology on $\mathbb{S}^1$ is just the induced topology by considering $\mathbb{S}^1$ as the circle of unit radius in $\mathbb{R}^2$, and the topology on $\mathbb{S}^1 \times \mathbb{S}^1$ is the product topology.

(i) Let $\psi : \mathbb{R}^2 \to \mathbb{S}^1 \times \mathbb{S}^1$ be the map defined by

$$\psi(x,y) = \left(e^{2\pi i x}, e^{2\pi i y}\right).$$

Show that ψ descends to a well-defined map $\bar{\psi} : \mathbb{R}^2/\mathbb{Z}^2 \to \mathbb{S}^1 \times \mathbb{S}^1$, such that $\psi = \bar{\psi} \circ p$. Show that $\bar{\psi}$ is a bijection.

$$\begin{array}{ccc} \mathbb{R}^2 & & \\ {\scriptstyle p}\downarrow & \searrow^{\psi} & \\ \mathbb{R}^2/\mathbb{Z}^2 & \xrightarrow[\bar{\psi}]{} & \mathbb{S}^1 \times \mathbb{S}^1 \end{array}$$

(ii) Use the definition of the quotient topology to check that p is continuous.

(iii) Prove that p is an open map. (For the definition of an open map, see Exercise 3 of Problem Sheet 4.)

[Hint: For $U \subset \mathbb{R}^2$, show that

$$p^{-1}(p(U)) = \bigcup_{a \in \mathbb{Z}^2} (a + U).$$

Here the notation is $a + U := \{a + x : x \in U\}$ - this is a translation of the set U by a.]

(iv) Use the definition of product topology to show that ψ is continuous. (You can use Exercise 3 of Problem Sheet 4.)

(v) Prove that ψ is an open map.

[Hint: Start with a base consisting of rectangles for the topology on $\mathbb{R}^2$.]

(vi) Deduce that $\bar{\psi}$ is both continuous and open, and hence a homeomorphism.

[Hint: Use that for $U \subset \mathbb{R}^2/\mathbb{Z}^2$, $p(p^{-1}(U)) = U$, and hence

$$\bar{\psi}(U) = \bar{\psi}\left(p\left(p^{-1}(U)\right)\right) = \psi\left(p^{-1}(U)\right).$$

Find a similar formula for $\bar{\psi}^{-1}(V)$ where $V \subset \mathbb{S}^1 \times \mathbb{S}^1$.]

Exercise 4. Recall that the real projective space $\mathbb{RP}^n$ is defined as the set of lines going through the origin:

$$\mathbb{RP}^n = \{L_x : x \in \mathbb{R}^{n+1} \setminus \{0\}\},$$

where L_x is the line passing through $0 \in \mathbb{R}^{n+1}$ and x, we can equip $\mathbb{RP}^n$ with the quotient topology $\mathcal{U}_p$, induced by the surjective map $p : \mathbb{R}^{n+1} \setminus \{0\} \to \mathbb{RP}^n$:

$$x \mapsto L_x.$$

Our goal is to show that $\mathbb{RP}^n$ is homeomorphic to $\mathbb{S}^n/\{\pm 1\}$.

(i) For every set $U \subset \mathbb{R}^{n+1}$, show that for every $y \in p(U)$, there exists $x \in U$, such that $L_y = L_x$.

(ii) Show that

$$p^{-1}(p(U)) = \bigcup_{\lambda \in \mathbb{R}^*} (\lambda U),$$

where $\lambda U = \{\lambda x : x \in U\}$ - this is the homothethy set of U by ratio λ).

(iii) Conclude that p is an open map.

(iv) By following a similar strategy as in Exercise 3, show that $\mathbb{RP}^n$ is homeomorphic to $\mathbb{S}^n/\{\pm 1\}$ via the following diagram.

$$\begin{array}{ccc} \mathbb{R}^{n+1} \setminus \{0\} & \xrightarrow{\quad p' \quad} & \mathbb{S}^n \\ \Big\downarrow{\scriptstyle p} & & \Big\downarrow{\scriptstyle \psi} \\ \mathbb{RP}^n & \xrightarrow[\quad \bar{\psi} \quad]{} & \mathbb{S}^n/\{\pm 1\} \end{array}$$

Exercise 5. Equip the set $\{0,1\}$ with the discrete topology, let $X = \{0,1\}^{\mathbb{N}}$, and consider the sequence $(e_n)_{n=1}^{\infty}$ in X where e_i has a 1 in position i and 0s everywhere else.

(i) Equip X with the box topology $\mathcal{U}_{\text{Box}}$, and prove that the sequence $(e_n)_{n=1}^{\infty}$ does not have a convergent subsequence.

(ii) Show that (e_n) converges in product topology.

Exercise 6. (Universal property for quotient topology.) Let $(X,\mathcal{U})$ be a topological space, and Y a set with a surjective map $p : X \to Y$. Prove that the quotient topology $\mathcal{U}_Y$ on Y is the unique topology of Y such that, for every topological space $(Z,\mathcal{V})$ and map $f : Y \to Z$, f is continuous if and only if $f \circ p : X \to Z$ is continuous.

